Mayar (May) **Ahmed**

Computational Drug Discovery | Structure Prediction, Docking & ML Scoring | Scalable Scientific Pipelines | Open to Relocation

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Summary

Computational scientist (PhD expected Apr 2027) with 5 years in structure-based drug discovery, building automated, scalable workflows across structure prediction, docking, and machine learning scoring models. Lead developer of **LIGNOVA** (258M+ quality-graded protein–ligand poses generated on HPC) and its async REST API client, **reqadence**.

Skills

ML Evaluation	Benchmarking vs. baselines, shortcut/failure-mode analysis, calibration, bootstrap tests; PyTorch, scikit-learn
Cheminformatics	RDKit, Open Babel, Gypsum-DL, Meeko; chemical dataset curation; similarity search; SAR/activity-cliff analysis
Modeling	Docking: Glide (Schrödinger), GNINA, Vina; co-folding (AlphaFold3); Foldseek; MD simulation, MDAnalysis
Software & HPC	Python, Bash, R; NumPy, pandas, Polars; Linux/SLURM (NSF ACCESS HPC); Git/GitHub Actions, pytest, pixi

Experience

Graduate Student Researcher, Dr. Jacob D. Durrant — *Pittsburgh, PA*

Department of Biological Sciences, University of Pittsburgh — *Sep 2021 – Present*

- Built **LIGNOVA**, a fully automated open-source pipeline that curates PubChem bioactivity and PDB data into ML training data: **258M+ poses** across 5.86M protein–ligand pairs (89% affinity-labeled), via **reqadence** ingestion and GNINA docking on HPC.
- Designed LIGNOVA's automated quality grading: calibrated CNN pose-score cutoffs on crystal-referenced complexes (redocking and cross-docking) so each **Gold/Silver/Bronze** tier meets 75% precision, then graded complexes without references.
- Benchmarked GNINA's CNN affinity model against **5M+ experimental drug–target labels**: it outranked Vina on 77% of ∼11K structures but barely beat a molecular-weight baseline, a **size shortcut** confirmed with permutation and bootstrap controls.
- Implemented **activity-cliff detection** in LIGNOVA (839K cliff pairs); on the docked subset, showed the CNN model is **cliff-blind**, predicting ∼0.25 pK gaps for 100-fold+ potency differences, with its edge over the MW baseline shrinking ∼60%.
- Led **structure-based reverse virtual screening** with the O'Donnell Lab for yeast arginine sensors missed by sequence homology (Foldseek, docking, AlphaFold3); **nominated 4 candidates**, a subset experimentally validated; **MD pose-stability runs ongoing.**
- Curated a benchmark of 215 X-ray complexes (1,415 ligands) across three hydration conditions, testing whether consensus rescoring (ComBind) could replace explicit-water modeling in **Glide** pose prediction. **Manuscript in preparation.**

Undergraduate Student Researcher, Dr. Eman Badr — *Giza, Egypt*

University of Science and Technology, Zewail City — *Jun 2020 – Apr 2021*

- Trained Random Forests (scikit-learn) on AML transcriptomes to predict drug sensitivity; identified genetic markers of response.

Education

University of Pittsburgh — *Pittsburgh, PA*

Ph.D. Candidate, Molecular, Cellular, and Developmental Biology — *Aug 2021 – Apr 2027 (expected)*

University of Science and Technology, Zewail City — *Giza, Egypt*

B.S., Computational Biology and Genomics, Biomedical Science Major; Cum Laude, Provost's Honors Roll — *Aug 2017 – Jul 2021*

Publications

Holland, M.*, **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, *122*(51). DOI: 10.1073/pnas.2520843122. (*co-first)

Kochnev, Y.; **Ahmed, M.**; Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, *52*(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, *21*(1), 299. DOI: 10.1186/s12915-023-01791-z.

Conference Posters & Seminar Talks

2026	**LIGNOVA: Large-Scale Generation of High-Quality Protein–Ligand Complexes**, Computational Medicinal Chemistry School	*Cambridge, MA*
2024	**Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery**, Comp. Chem. GRC	*Portland, ME*
2023	**Augmenting Protein–Ligand Complex Databases with PubChem**, CADD GRC	*West Dover, VT*
2022–26	**4 talks** on LIGNOVA & its application in fragment-based ML design, Pitt MCDB Noon Seminar Series	*Pittsburgh, PA*

Awards & Mentoring

2026	**NSF ACCESS Allocation (BIO260240)**, Purdue Anvil & TAMU ACES compute to scale LIGNOVA	*USA*
2024–25	**Biochemistry TA & Fellow**, ∼200 students; mentored 15 undergrad TAs; Baranger Teaching Award nominee	*Pittsburgh, PA*
2024	**Research Mentor, TECBio REU**, co-mentored an undergraduate AlphaFold2 drug-design project	*Pittsburgh, PA*